

distribution compared to that of structures generated from rattled relaxations and rattled Boltzmann sampling. We also observe a shift to lower energies for structures generated using AIMD.

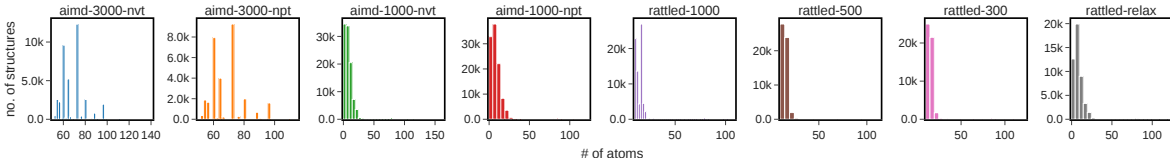


Figure B.1 Distribution of atoms per structure across OMat24. Histogram of number of atoms per structure per sub-dataset in OMat-24 dataset.

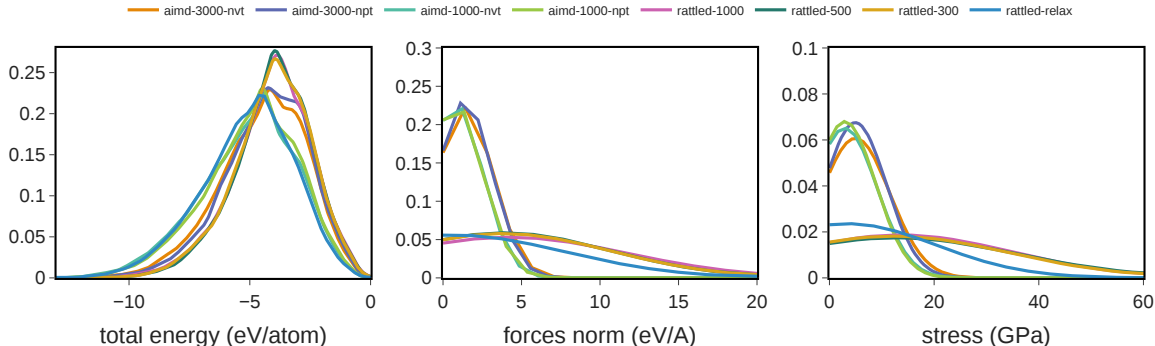


Figure B.2 Density distributions across OMat24 sub-datasets. Density distributions showing energy, forces norm and maximum absolute stress densities for all sub-datasets in OMat-24.

Figures B.3 and B.4 show per element histograms of the number of occurrences in each dataset. The histograms are created by randomly sampling 1% and 10% of each dataset respectively, as a result elements that appear very rarely may have been missed. By comparing Figures B.3 and B.4 with the corresponding histogram for the OMat24 dataset in Figure 1c we observe that the distributions between OMat24 and Alex are very similar and more uniformly distributed than that of MPtrj.

The composition and structural diversity of the OMat24 dataset are further demonstrated by examining the variety of local atomic neighborhoods. Figure B.5 displays the total number of distinct element pairs within a 3.5Å radius for the OMat24, Alexandria, and MPtrj datasets. The pair counts for Alexandria and OMat24 are several orders of magnitude greater than those for MPtrj, reflecting the relative sizes of the datasets. Furthermore, the OMat24 dataset exhibits more uniform and comprehensive coverage of near-neighbor pairs compared to both Alexandria and MPtrj.

B.1 Drift forces in AIMD data

Nonnegligible drift forces—also referred to as nonzero net forces—can arise in Ab Initio Molecular Dynamics (AIMD) calculations. VASP routines are written to correct the drift and ensure dynamics consistent with the ensemble being used. However, the forces printed in output files of AIMD calculations include the drift terms. A recent study found that AIMD data in the OMat24 dataset included several structures with substantial drift forces (up to 1 eV/Å)⁴⁴. Figure B.6 presents the mean net force per atom for AIMD subsplits in the OMat24-1M test set. Consistent with the independent study⁴⁴, net forces can reach up to 1 meV/Å in AIMD calculations when using OMat24 settings. We also recomputed the data as single point calculations of the AIMD frames and with a much stricter electronic convergence (EDIFF=1 × 10⁻⁶), and observe that the net forces are no longer present.

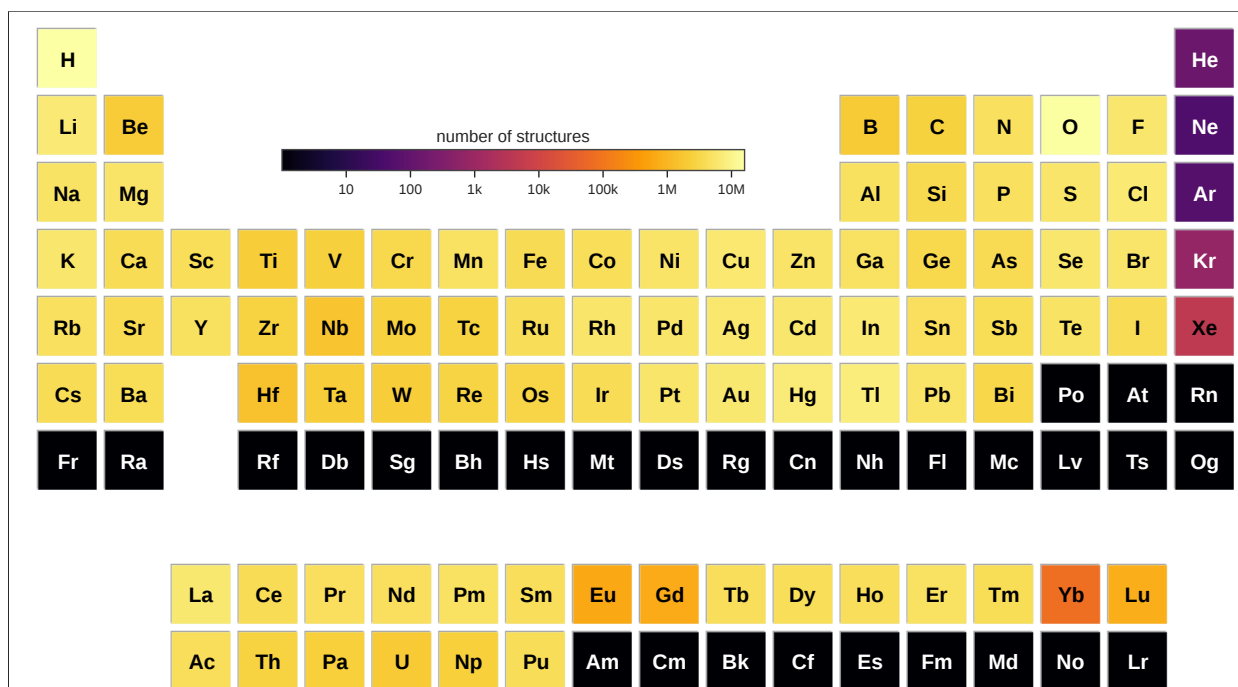


Figure B.3 Element distribution in Alexandria PBE dataset.²⁰

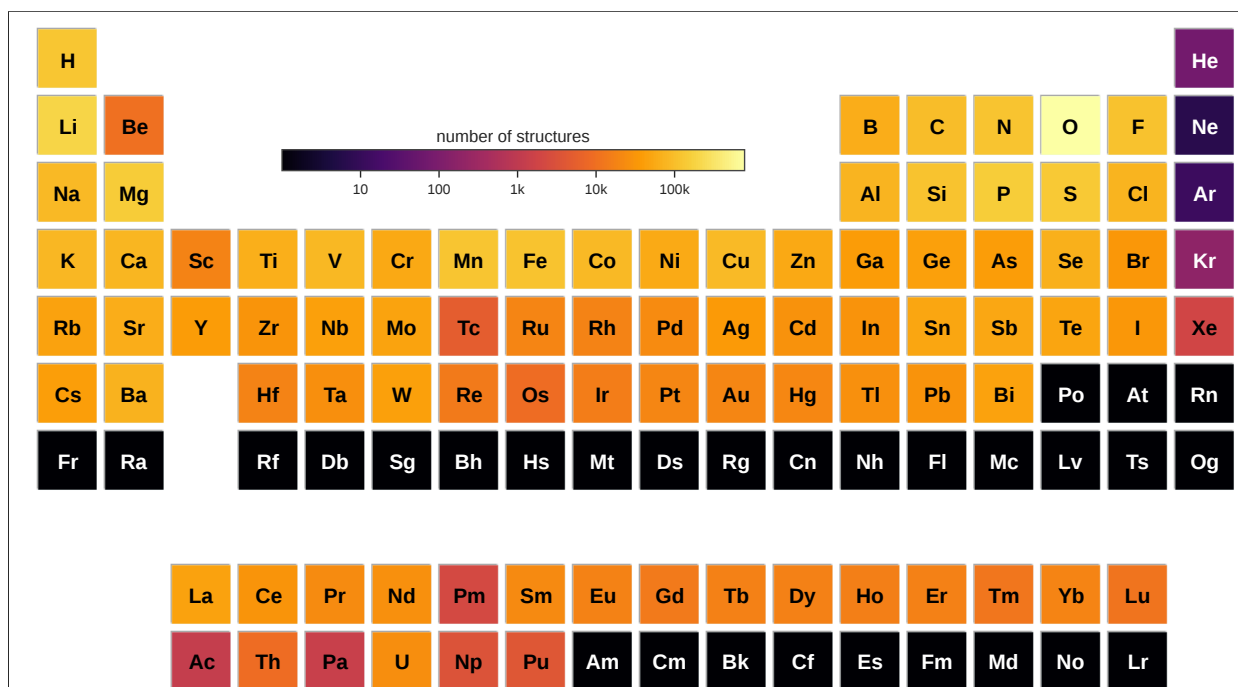


Figure B.4 Element distribution in MPtrj dataset.¹⁶

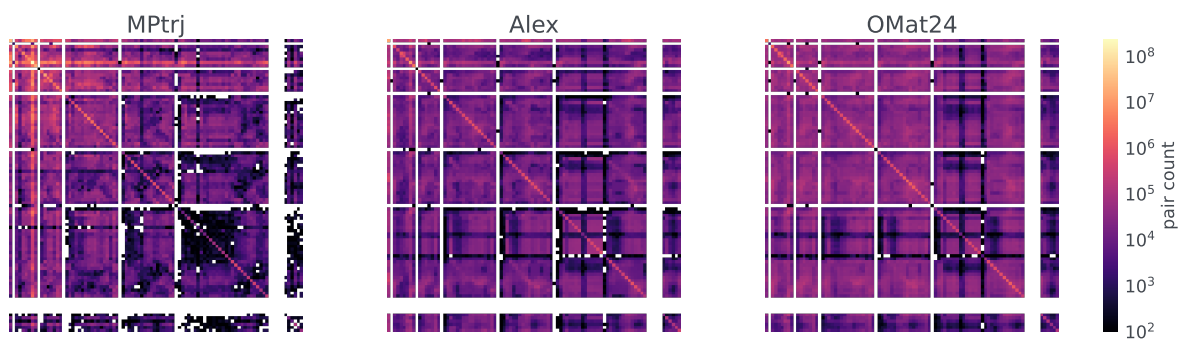


Figure B.5 Number of distinct pair interactions for atoms within 3.5 Å.

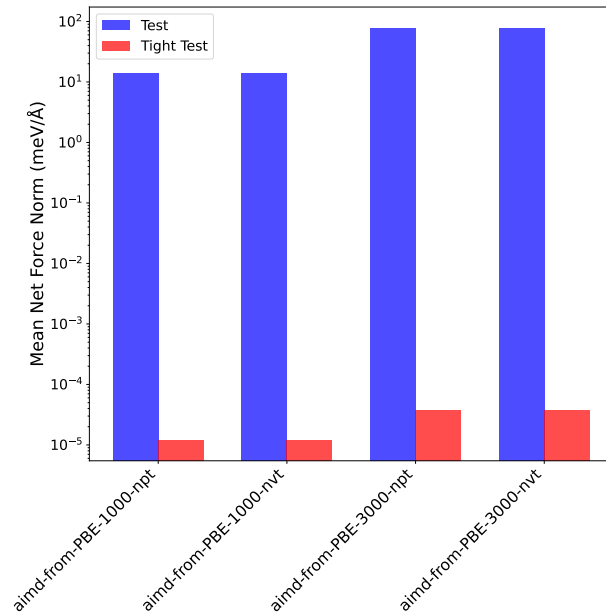


Figure B.6 Mean drift forces in AIMD. Mean drift (net forces) per atom for AIMD 1M test subsplits calculated with different energy convergence thresholds. (Test) OMat24 threshold, (Tight Test) stricter 1×10^{-6} threshold.

Following similar analyses⁸¹, we find that machine-learned interatomic potentials (MLIPs) trained on OMat24 as-is predict tightly converged DFT forces with accuracy comparable to, or even better than, predictions for less tightly converged calculations. Table B.1 presents the test MAE for eSEN and eqV2-S models on both the regularly converged and tightly converged test sets, showing that errors are effectively equivalent. This result aligns with previous experiments and suggests that MLIP models can learn the true forces without being affected by this level of numerical noise from drift forces. We do not advocate for future data generation with looser convergence thresholds. Rather, future efforts should aim to balance computational efficiency with fully converged calculations considering the effects of artifacts such as drift-forces^{58,81,82}.

Table B.1 Model prediction performance vs DFT convergence. Model prediction MAE over a subsample of the test dataset and the same data points computed tighter convergence settings.

Model	Energy MAE (meV/atom)		Force MAE (meV/Å)	
	Regular	Tight	Regular	Tight
eSEN	9.0	6.9	40.1	37.9
eqV2 S	10.0	8.1	41.9	39.9

C OMat24 elemental references and formation energy corrections

In order to calculate formation energies and energies above hull involving DFT calculations using GGA and GGA with Hubbard U corrections, the Materials Project has developed a mixing compatibility correction⁴¹. In addition, an energy adjustment to certain anion species is added to account for electron localization errors⁴¹. These compatibility corrections are obtained by way of a least squares fit minimizing the error between predicted formation energies and a set of experimental formation energies with reported uncertainties below 100 meV/atom⁴¹.

We have followed the same procedure and computed a compatibility correction set for OMat24 DFT settings. We use the same 222 compounds predicted to be within 100 meV of the MP convex hull that are used in fitting the MP2020 Compatibility corrections. The +U mixing and anion correction values per element/species are listed in Tables C.1 and C.2 respectively.

Table C.1 GGA/GGA+U transition metal mixing corrections. Fitted GGA+U mixing energy corrections and their associated uncertainties for transition metal oxides and fluorides (in eV/atom). The corrections were fit following the MP2020 Compatibility scheme⁴¹.

Species	OMat24 Correction	OMat24 Uncertainty	MP Correction	MP Uncertainty
V	-1.813	0.0064	-1.7	0.0064
Cr	-2.037	0.0108	-1.999	0.0108
Mn	-1.701	0.0053	-1.668	0.0053
Fe	-2.428	0.0101	-2.256	0.0101
Co	-2.151	0.006	-1.638	0.006
Ni	-2.58	0.0107	-2.541	0.0107
W	-4.445	0.0253	-4.438	0.0253
Mo	-2.972	0.0089	-3.202	0.0089

Table C.2 GGA/GGA+U anion mixing corrections. Fitted Composition-based energy corrections and uncertainties for various anions (in eV/atom) . The corrections were fit following the MP2020 Compatibility scheme⁴¹

Anion/Element	OMat24 Correction	OMat24 Uncertainty	MP Correction	MP Uncertainty
oxide	-0.657	0.002	-0.687	0.002
peroxide	-0.433	0.0172	-0.465	0.0172
superoxide	-0.152	0.0075	-0.161	0.0075
S	-0.487	0.0093	-0.503	0.0093
F	-0.436	0.0026	-0.462	0.0026
Cl	-0.6	0.0018	-0.614	0.0018
Br	-0.318	0.0026	-0.534	0.0026
I	-0.194	0.0055	-0.379	0.0055
N	-0.303	0.0093	-0.361	0.0093
Se	-0.474	0.0341	-0.472	0.0341
Si	0.028	0.0165	0.071	0.0165
Sb	-0.194	0.0089	-0.192	0.0089
Te	-0.418	0.0262	-0.422	0.0262
H	-0.173	0.0013	-0.179	0.0013
ozonide	0	0	0	0

The parity plot of the DFT predicted formation energies and experimental formation energies is shown in Figure C.1. The resulting mean absolute error between the DFT predicted and experimental formation energies for the 222 compounds is 55 meV/atom. Figure C.2 shows the difference in predicted and experimental formation energy for compounds with the largest experimental formation energy uncertainties (> 10 meV/atom).

In order compute formation energies using OMat24 DFT settings or models trained on OMat24 only, we have also computed the energy of elemental reference compounds. Figure C.3 shows the energy of elemental reference compounds computed with the OMat24 DFT settings compared to the PBE calculations in MP. We